

Cox-Ingersoll-Ross (CIR) Process Age Inference Algorithm for Age Estimation Manual (v.1.0)

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1. Introduction

The MATLAB software for the Cox-Ingersoll-Ross (CIR) Process Age Inference Algorithm is designed to construct radiocarbon age models, benthic $\delta^{18}\text{O}$ -aligned age models, and multiproxy age-depth models (age models that leverage age information from both dating techniques). We regard this software as the first half of the next version of BIGMACS [Lee and Rand et al. (2023), “Bayesian age models and stacks: combining age inferences from radiocarbon and benthic $\delta^{18}\text{O}$ stratigraphic alignment”, *Climate of the Past*]: the second half will be about the stack construction, which will be available soon.

Here we provide instructions to download the software, prepare the required input files, construct age-depth models, and analyze the results. It is our hope that this software provides its users with a less subjective and more realistic method to construct ocean sediment core age models.

When you are using this software, please cite the following paper:

Lee et al. (2026), “Scalable and consistent sedimentation rate modeling via the Cox-Ingersoll-Ross (CIR) process for ocean sediment core age model development”, in review and submitted to *Geoscientific Model Development*.

Happy chronology building! We also provide a **Python** version at https://github.com/eilion/CIR_Process_Aligner.

2. Installation

The latest version of the open-sourced software package can be installed from GitHub using the following link:

https://github.com/eilion/CIR_Process_Aligner_MATLAB/Aligner_MATLAB

The software was developed and tested on MATLAB R2024a and requires the statistics and machine learning toolbox and the parallel computing toolbox. An example run is included with the installation. To confirm that the software is running properly, open and run the script `'/main.m'`. This will begin a multiproxy alignment for the example cores GeoB9508-5, GIK13289-2, MD95-2042 (up to 18 meters) and ODP658C to the DNEA stack [Lee and Rand et al., 2023].

3. Preparing input files

To construct age-depth models, create a new folder as `/Inputs/[inputFolder]` with the desired run name. All proxy data files should be of the txt format. See one of the input folders as an example for structure and formatting. Under `/Inputs/[inputFolder]/Records/`, make folders of records, and, under each subfolder `/Inputs/[inputFolder]/Records/[core_name]/`, put the following txt files:

3.1. C14_data.txt stores the radiocarbon and calibration data for the sediment core and has the following six columns: (1) depth (in meters); (2) age (in kiloyears); (3) error (in kiloyears); (4) dR (in kiloyears); (5) dSTD (in kiloyears); and (6) cc ('1' for IntCal20, '2' for Marine20, and '3' for SHCal20).

3.2. d18O_data.txt stores the $\delta^{18}\text{O}$ data for the sediment core and has two columns: (1) depth (in meters); and (2) $\delta^{18}\text{O}$ (in permils).

3.3. additional_ages.txt stores any additional age information (e.g., tephra layers, tie points, magnetic reversals, etc.) and has three columns: (1) depth (in meters); (2) age (in kiloyears); and (3) unct (in kiloyears). Age information is fed as a Gaussian distribution with mean 'age' and standard deviation 'unct'.

3.4. record_info.txt specifies the following four information to run the core:

- **num_Z** is the number of inducing depths. If it is set to be NaN, then this value is specified as the number of proxy observations.
- **init_sedrate** initializes the standardization parameter.
- **init_scale** and **init_shift** initialize the $\delta^{18}\text{O}$ scale and shift parameters, respectively.

If a core does not have a certain proxy type, you do not need to put the corresponding txt file in the input folder.

For the stack to be aligned, put stack.csv file as `/Inputs/[inputFolder]/stack.csv`: see `/Inputs/EXPERIMENT/stack.csv` as an example. It consists of three columns: (1) age (in kiloyears); (2) mean (in kiloyears); and (3) sigma (in kiloyears). The format corresponds to stack files in BIGMACS.

To configure the alignment setting for each record, you need to first copy and paste `/Defaults/setting_record.txt` to `/Inputs/[inputFolder]/Records/[core_name]/setting_record.txt` and then modify the values in the txt file as you want:

- **nSamples** is the number of age-depth model samples to draw.
- **islearn_average_sedrate** determines whether the standardization parameter will be updated ('Yes') or not ('No'). If it is set to be 'No', then the parameter will be what is specified as 'init_sedrate' in 'record_info.txt'.
- **islearn_scale** determines whether the $\delta^{18}\text{O}$ scale parameter will be updated ('Yes') or not ('No'). If it is set to be 'No', then the parameter will be what is specified as 'init_scale' in 'record_info.txt'.

- **islearn_shift** determines whether the $\delta^{18}\text{O}$ shift parameter will be updated ('Yes') or not ('No'). If it is set to be 'No', then the parameter will be what is specified as 'init_shift' in 'record_info.txt'.
- **alpha, beta, rho** are values of the CIR-process model parameters in the manuscript.
- **a_d18O, b_d18O, a_14C, b_14C** are the parameter values of the generalized Student's t-distributions for ^{14}C and benthic $\delta^{18}\text{O}$.
- **start_depth** and **min_start_age** specify the start depth in meters and the minimum age threshold in kiloyears at that depth.

4. Running the algorithm

To run the algorithm, just execute the command `Aligner([inputFolder])`, where [inputFolder] is the name of the folder in /Inputs. For example, execute the command `Aligner('EXPERIMENT')`. See `\main.m` for more details.

5. Output files

All output files will be saved in a mat file under 'Outputs/[inputFolder]/Trial_XXX/' consisting of five structs.

5.1. C14_CI contains the pointwise ^{14}C age estimation. `C14_CI.depth`, `C14_CI.lower`, `C14_CI.median`, `C14_CI.upper` and `C14_CI.samples` consist of the depths of ^{14}C proxy observations, lower bounds of the 95% credible intervals of ages, medians of ages, upper bounds of the 95% credible intervals of ages, and age samples, respectively. These are used only for visualization (blue bars in the first panel) by running `/getFigures.m`. Do NOT use these estimates as the output of this software.

5.2. data contains the input information. `data.depth` consists of the unique depths of proxy observations; `data.d18O` consists of the $\delta^{18}\text{O}$ values and their location allocations of `data.depth`; `data.C14` consists of the ^{14}C data with the last column for their location allocations of `data.depth`; `data.ETC` consists of the additional age information with the last column for their location allocations of `data.depth`; `data.scale` and `data.shift` are the estimated $\delta^{18}\text{O}$ scale and shift parameters, respectively; `data.R` is the estimated standardization parameter; and `data.num_Z` is the number of inducing depths.

5.3. SAMPLES contains the raw samples of sedimentation rates and age-depth models. `SAMPLES.ACC_RATE` consists of the *inverse* sedimentation rate samples at depths `SAMPLES.D`, column-by-column; `SAMPLES.AGE_D` consists of the age-depth model samples at depths `SAMPLES.D0`. This struct contains the output of this software.

5.4. setting represents the input settings.

5.5. target stores the benthic $\delta^{18}\text{O}$ stack and ^{14}C calibration models.

In addition, we provide a function 'getFigures.m' to visualize the results based on the mat file. To run it, execute the command `/getFigures('inputFolder/Trial_XXX')`. For example, `getFigures('EXPERIMENT/Trial_001')`. The path indicates where the mat file is stored under 'Outputs'.